

SESSION 7 • LECTURE 6

First Law of Thermodynamics — Closed Systems

Detailed lecture notes — Joule's experiment, statements & proof, internal energy, specific heats, PMM1, solved numericals

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Learning objectives. After this session you should be able to: (i) describe Joule's experiment and its conclusion; (ii) state the first law for a cycle and for a process; (iii) show that $(Q - W)$ defines the property E ; (iv) use $Q - W = \Delta U$ with correct signs; (v) define c_v and c_p and use $\Delta U = mc_v\Delta T$ for ideal gases; (vi) explain why PMM1 is impossible.

1. Joule's Paddle-Wheel Experiment (1843-48)

1. An insulated tank of water carries a paddle wheel driven by a falling weight; the measured work input is $W = (\text{weight}) \times (\text{drop height})$.
2. Stirring raises the water temperature — the system changes state by pure work input.
3. The insulation is then removed and the water is cooled back to its initial state; the heat rejected Q is measured.
4. Finding: for a system executing a cycle, the ratio W/Q is always the same constant, regardless of the amounts or the type of work. In SI units both are joules, and net work input equals net heat output.

Conclusion: heat and work are mutually convertible forms of the same quantity — energy. Joule's constant, 4.186 kJ/kcal, converts the old heat unit to the work unit.

2. Statements of the First Law

$$\oint \delta Q = \oint \delta W$$

For a closed system executing a cycle: net heat = net work

$$Q_{1-2} - W_{1-2} = \Delta E = E_2 - E_1$$

For a process: energy supplied minus energy delivered is stored

Proof that $(Q - W)$ defines a property

1. Let two different processes A and B connect states 1 and 2. Consider the cycle $1 \rightarrow 2$ (via A) $\rightarrow 1$ (via reversed C), and similarly with B and C.
2. Applying the cyclic first law to both cycles and subtracting: $(Q - W)_A = (Q - W)_B$.
3. Thus $(Q - W)$ has the same value for every path between 1 and 2 — it depends only on the end states, and therefore defines a property: the total energy E .
4. $E = U + KE + PE$. For stationary closed systems $\Delta KE = \Delta PE = 0$, giving the working form **$Q - W = \Delta U$** .

KEY POINT Energy can be neither created nor destroyed in a process; it can only change form. Individually Q and W are path functions, but their difference is a point function.

3. Internal Energy and Specific Heats

- **Internal energy U :** the sum of all microscopic energies (molecular translation, rotation, vibration, intermolecular potential). Extensive; $u = U/m$ in kJ/kg.
- **$c_v = (\partial u / \partial T)_v$.** At constant volume, $W = 0$ and $Q = \Delta U = mc_v\Delta T$.
- **$c_p = (\partial h / \partial T)_p$.** At constant pressure, $Q = mc_p\Delta T$; the excess over ΔU supplies the boundary work $p\Delta V = mR\Delta T$.
- For an ideal gas: $c_p - c_v = R$ and $\gamma = c_p/c_v$. Air: $c_p = 1.005$, $c_v = 0.718$, $R = 0.287$ kJ/kg·K, $\gamma = 1.4$.
- **Joule's law:** for an ideal gas $u = u(T)$ only; hence $\Delta U = mc_v\Delta T$ for *any* process, not merely constant volume.

4. Perpetual Motion Machine of the First Kind (PMM1)

A PMM1 would deliver work continuously while absorbing no energy from any source — a cycle with $W_{\text{net}} > 0$ and $Q_{\text{net}} = 0$, violating $\oint \delta Q = \oint \delta W$. The first law declares it impossible; equally, a machine cannot continuously destroy energy. Every "free energy" claim fails an honest energy audit.

5. Worked Examples

WORKED EXAMPLE 6.1 — Process and adiabatic restoration

During a process, 80 kJ of heat is supplied to a closed system while it does 30 kJ of work. Find ΔU . If the system is then restored to its initial state adiabatically, find the work in the return process.

Solution.

- 1→2: $Q = +80 \text{ kJ}$, $W = +30 \text{ kJ} \Rightarrow \Delta U_{12} = 80 - 30 = +50 \text{ kJ}$.
- 2→1 adiabatic: $Q = 0$; $\Delta U_{21} = U_1 - U_2 = -50 \text{ kJ}$.
- 0 - $W_{21} = -50 \Rightarrow W_{21} = +50 \text{ kJ}$ (work done by the system).
4. The 50 kJ stored in 1→2 leaves entirely as work in the return.

ANSWER $\Delta U_{12} = +50 \text{ kJ}$ | $W_{21} = +50 \text{ kJ}$

WORKED EXAMPLE 6.2 — Compression with heat rejection

A gas is compressed with 95 kJ of work done on it while 43 kJ of heat is rejected. Find ΔU .

Solution.

1. Signs: $W = -95 \text{ kJ}$ (on the system), $Q = -43 \text{ kJ}$ (rejected).
2. $\Delta U = Q - W = (-43) - (-95) = +52 \text{ kJ}$.
3. Internal energy rises: compression stores more energy than the cooling removes.

ANSWER $\Delta U = +52 \text{ kJ}$

WORKED EXAMPLE 6.3 — Constant-volume heating

1.5 kg of air ($c_v = 0.718 \text{ kJ/kg}\cdot\text{K}$) in a rigid tank is heated from 300 K to 420 K. Find Q , W and ΔU .

Solution.

1. Rigid tank: $V = C \Rightarrow W = 0$.
2. $\Delta U = mc_v\Delta T = 1.5 \times 0.718 \times 120 = 129.2 \text{ kJ}$.
3. $Q = \Delta U + W = 129.2 + 0 = 129.2 \text{ kJ}$.

ANSWER $Q = \Delta U = 129.2 \text{ kJ}$; $W = 0$

WORKED EXAMPLE 6.4 — Electric heater in a closed room (energy audit)

A sealed, insulated room receives 1.5 kW from an electric heater for 20 minutes. Treating the room air as the closed system, find the increase in its internal energy.

Solution.

1. Electrical work crosses the boundary INTO the system: $W = -(1.5 \times 20 \times 60) \text{ kJ} = -1800 \text{ kJ}$; $Q = 0$ (insulated).
2. $\Delta U = Q - W = 0 - (-1800) = +1800 \text{ kJ}$.
3. All electrical work becomes stored internal energy of the air.

ANSWER $\Delta U = +1800 \text{ kJ}$ (1.8 MJ)

6. Practice Problems (answers given)

#	Problem	Answer
P1	A system receives 120 kJ of heat and 40 kJ of paddle work while doing 65 kJ of expansion work. Find ΔU .	+95 kJ
P2	In an adiabatic expansion a gas does 75 kJ of work. Find ΔU .	-75 kJ
P3	2 kg of air is cooled at constant volume from 500 K to 350 K. Find Q .	-215.4 kJ
P4	True/false: "A system contains 100 kJ of heat." Justify.	False — heat exists only in transit

7. Summary

- Cycle: $\oint \delta Q = \oint \delta W$. Process: $Q - W = \Delta E$; stationary systems: $Q - W = \Delta U$.
- $(Q - W)$ is path-independent and defines the property E ; U is its microscopic part.
- $c_v = (\partial u / \partial T)_v$, $c_p = (\partial h / \partial T)_p$; ideal gas: $c_p - c_v = R$, $\Delta U = mc_v \Delta T$ for any process.
- PMM1 (work from nothing) violates the first law and is impossible.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 4 (First Law of Thermodynamics). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.